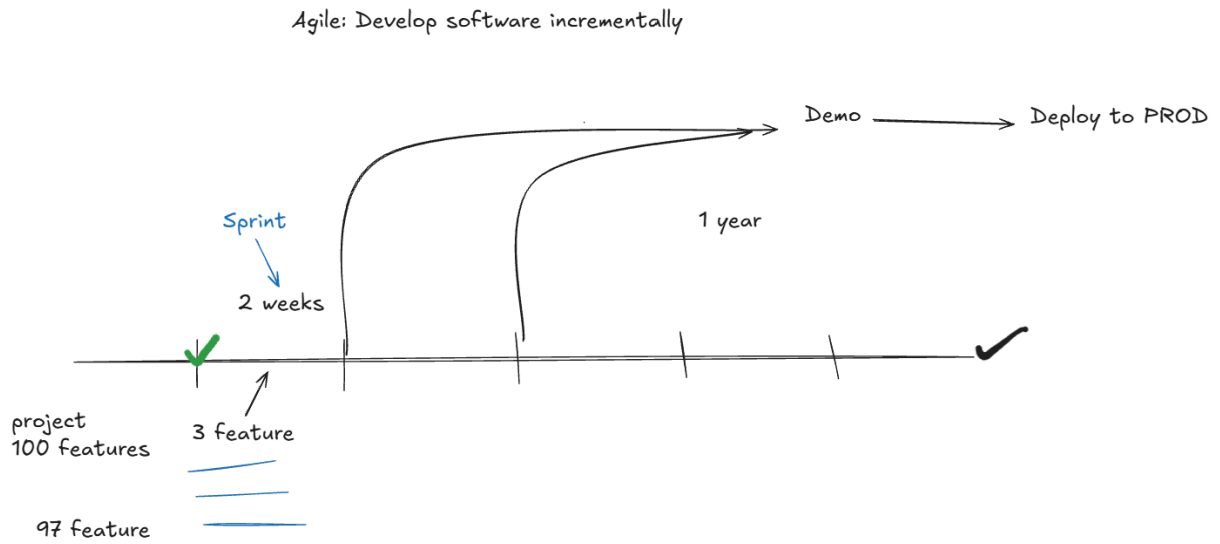
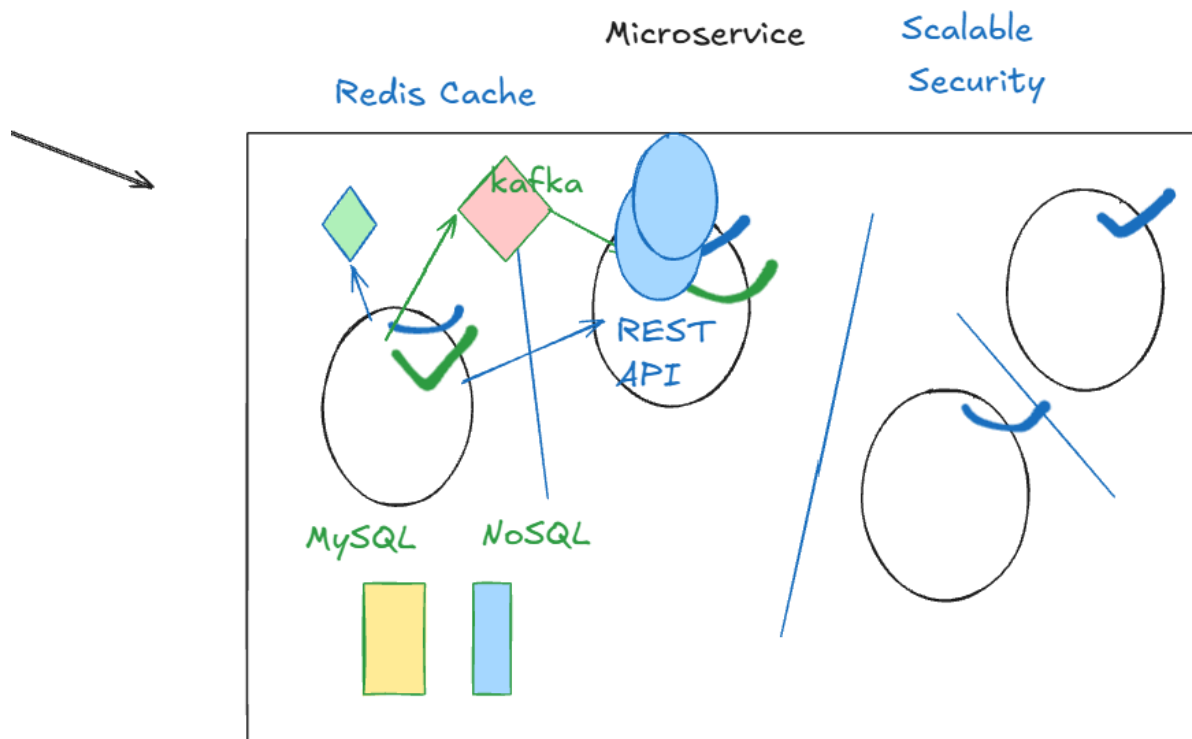


LIVE NOTES

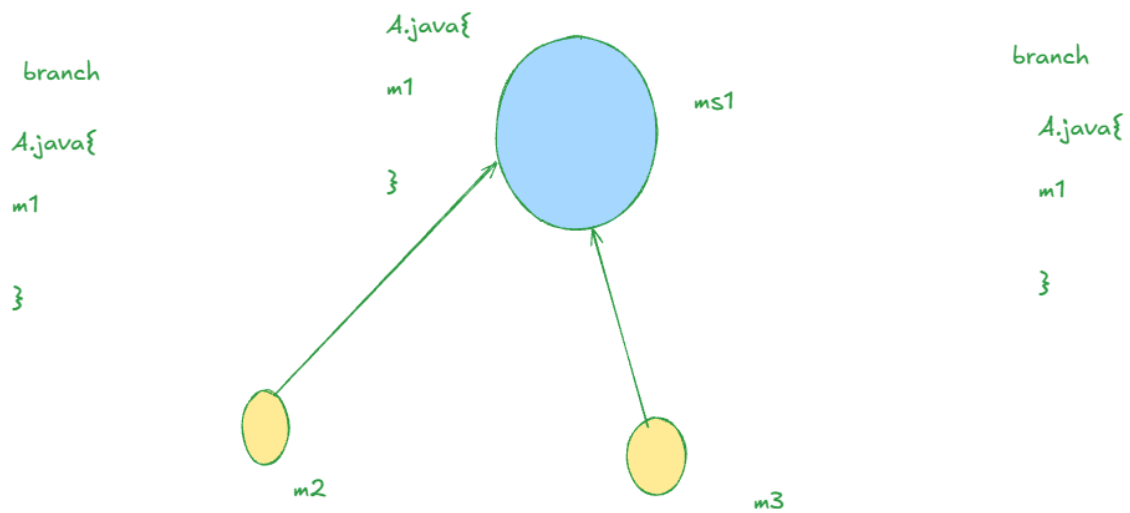
Agile software development



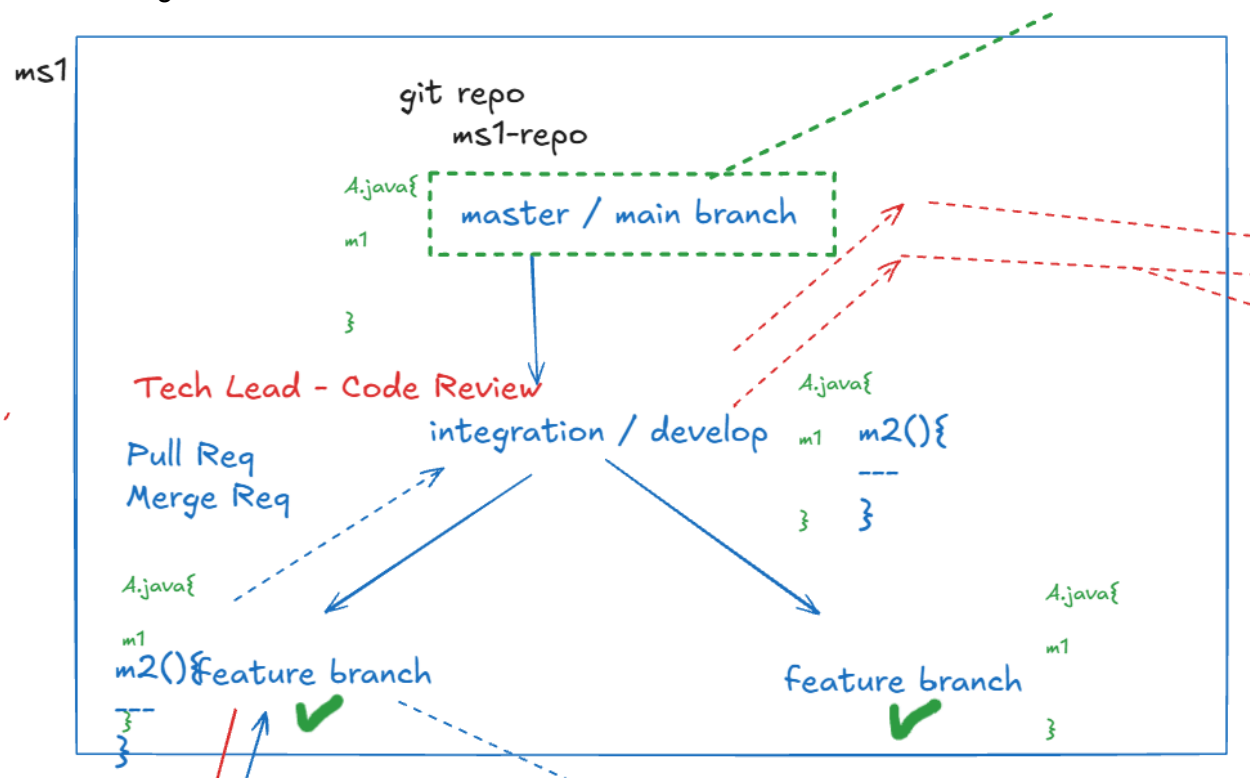
Architecture of the project



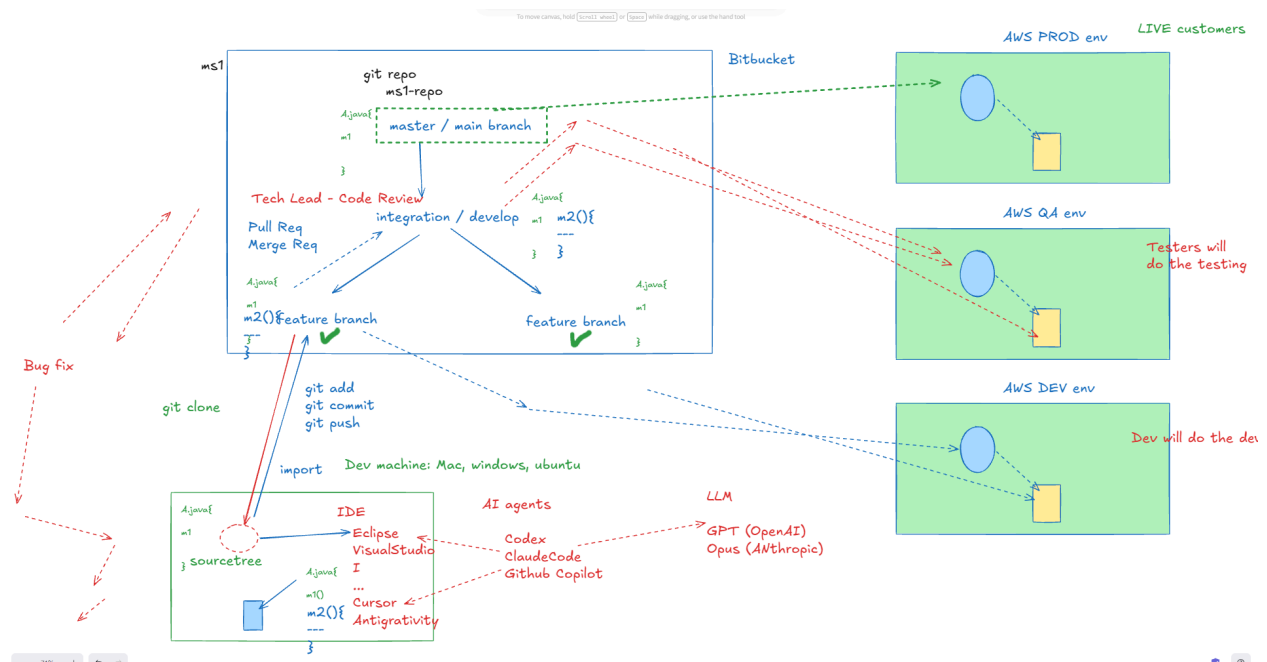
Branches are copy of code



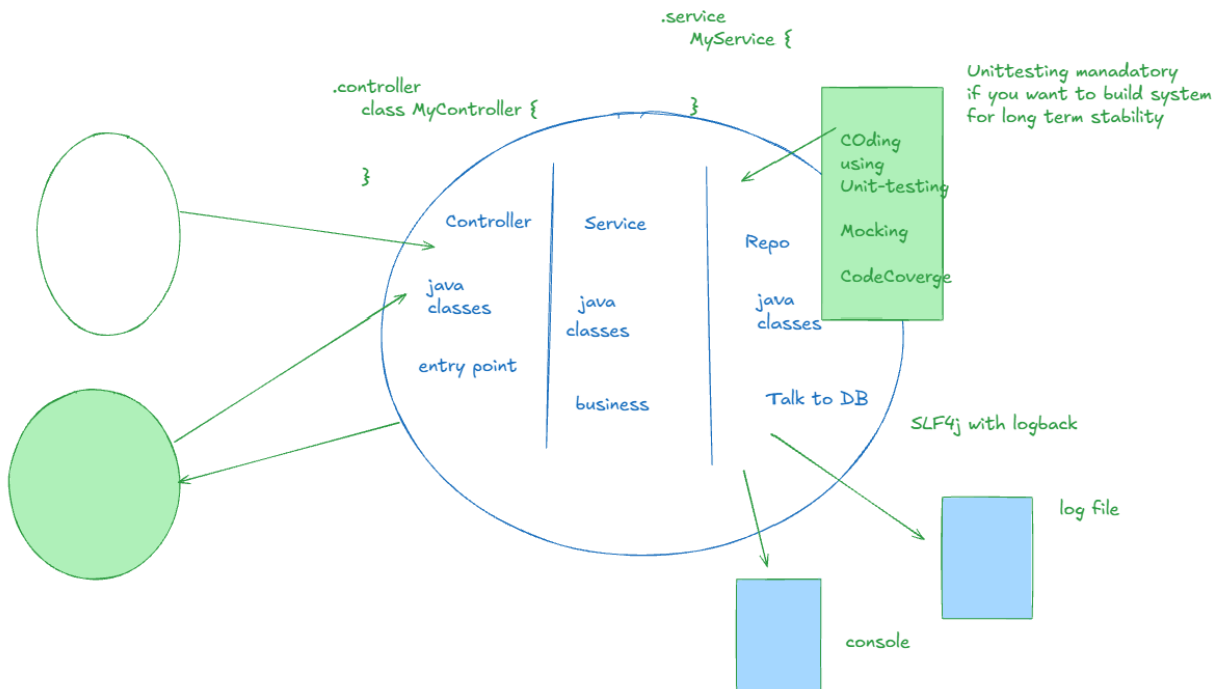
Git branching



End-to-end working with Git.



Layered app, invokers calling, logging, unittesting



Tausief
15+ yrs of IT exp
2011

Java dev
full stack
tech lead - 4/5team
Manager - 10-15mem
Senior manager - 25+ members
Head of Tech - 40+ members

Startups, Service, Product
High pace demanding client

10+yrs interview candidates
multiple campus placement

Every year, hikes, promotion,

0-3 yrs of exp - 2months Java Trainee Program.

=====

Your current knowledge will grow by 80%
You would gain knowledge equal to 2-3yrs of working exp.

=====

Mon-Fri, LIVE of 3hrs.
7 Week of live
100+hrs LIVE

Daily tasks

You would need to put additional 4-5hrs

My belief
You are smart. Willing to work hard
Right direction is missing

Help you in getting job quickly & building a successful carrer.

=====

I AM SMART, I AM HARD WORKER, WITH RIGHT DIRECTION, I WILL STRONGLY WORK
ON MY DREAMS, AND GET JOB & BUILD A GREAT SUCCESFUL CAREER FOR MYSELF

=====

YOU SHOULD GET JOB & MAKE A GREAT CARRER

Strong Preparation

- Core (Apti + DSA)
- Framework - Spring & Spring Boot
- Project
- Coding
- Communication
- Modern AI practices

Strong Jobhunting

- Resume
- skills
- LinkedIn
- GitHub
- Apply
- network
- events
- follow
- referral
- keep updating your profile

Consistently

5 week - recording..

<https://www.tausiefs.com/learn/JobSeeker-Daily-Standup->

recording - notes - tasks

<https://www.whatsapp.com/channel/0029Va4qa6CBFLgLqIw5g73I>

=====

Strong Realtime project development

- Framework - from project perspective
- Coding + System design from project

If you don't have strong realtime project
they don't shortlist your resume

Internships, Apprentice, Trainee program. build realtime project
Self project - Industry standards.

The way work happens in the industry. tools, process, technology

<https://roadmap.sh/backend>

=====

Building a realtime project, includes, knowing how the industry operates, tools, processes, technologies, frameworks, concepts, end-to-end app development.

=====

2 months Java Trainee Program.

100+ hrs of LIVE Realtime project development using Spring Boot & Microservice.

Pre: Strong Core Java + JDBC

Tech Skills

=====

✓ Programming Languages: Java

✓ Frameworks: Spring Boot, Spring Web, Spring Validation, Spring JDBC, Spring

Security

✓ Microservices Architecture, Circuit Breaker, Eureka Service Registry, Spring Cloud

Config

✓ AI Assisted Coding: GitHub Copilot, Claude, ChatGPT, Automated test generation,

API documentation

✓ RESTful APIs: Resource Urls, JSON, cURL, Postman, Swagger

✓ Databases: MySQL (AWS RDS)

✓ Version Control Systems: Git, Bitbucket, SourceTree

✓ Build Tools: Maven

✓ Server: Embedded Tomcat

✓ Testing Frameworks: Unit Testing (JUnit), Mocking (Mockito), Code Coverage

- ✓ Cloud Platforms: AWS (Amazon Web Services) - EC2, RDS, Secrets Manager
- ✓ IDEs: Eclipse, DBeaver
- ✓ Cache Technologies: Redis
- ✓ Agile Methodologies: Scrum, Agile Development
- ✓ Project management tool: Jira
- ✓ Logging, Debugging & Troubleshooting: Eclipse, Slf4j with LogBack
- ✓ Security and Authentication: Spring Security, HMAC SHA256, Basic Auth
- ✓ Performance Tuning | Design Patterns | Code Review | Data Structures | Algorithms
- ✓ Other tools: Jackson, Lombok, SonarQube, MobaXterm

=

====

Along with your core coding knowledge, I should also develop skills needed to work on realtime systems.

Add these skills to resume

Present the project confidently during interview.

====

How IT industry works for Java project development.

=====

Tools & process

=====

SDLC - Software development Life cycle

PLC

From start to end of the project, what does the company do.

AGILE - SDLC

How to implement AGILE in your project

SCRUM, Kanban, ScrumBan

Agile - Scrum methodology

what is Agile

Roles in Agile

Product Owner, Scrum Master, Team

- Product Owner - give you requirement
- Scrum Master - to ensure all sprints happen smoothly
wherever there is problem, SM works to unlock
- Team
 - Arch + Highlevel Design, + low level + coding + AI tools + Unittesting + QA testing.

How the work / features are given for development.

Project Management Tool: Jira

Why is Jira used in industry?

PO, Product - logs in to Jira

goes to backlog

creates a user-story(requirement will be written in the form user-story)

Add description

assigns the user-story/ticket to himself

=====

Grooming meeting / refinement / requirement gathering meeting

PO + SM + Tech

the requirements will be discussed with the tech team.

Tech team raises questions, PO is supposed to clarify

PO hand's over the user-story to tech members.

=====

PO will ask the tech team, to give story point estimate

fibonnaci number

1 2 3 5 8 ...

====

Tech team

internal discussion on how to implement this feature
architecture + highlevel system design + lowlevel system design

architecture + highlevel system design
Microservice
how many microservice

Microservice

don't code everything as one big project.
rather code them as separate projects, which are services.
for this feature, which-which microservice you will do coding into

architecture + highlevel system design

once these discussions are done, then story point estimate will be given to the user-story

=====

=====

1. PO write requirements in Jira usestory
2. discusses with tech team
3. Tech Team, finalize architecture - highlevel design - tech implementation approach
4. Gives story point estimate

=====

Sprint Planning Meeting

- Day1 of the sprint
- team will finalize how many features they can do in that sprint.
- work will be assigned, which developer will work on which feature

=====

IT has a process for coding.

- For the feature, we want to do coding in ms1 (microservice)
- Code for this microservice (ms1) will be maintained in Git repository
every single line of code that you do, will be tracked.
who made this change, what was the change, what date the change was made.

version control system => Git

=====

=> The code of the project, should be maintained in Git, so it can be tracked, who is making what changes.

=====

same microservice, same java class, and parallel 10 devs want to modify the code

10 devs should access same code, so code should be available in central place

====

1. Code should be kept git repo
2. Keep this git repo in cloud, central place, wherever everyone can access it.
3. Git Hosting provider
 - GitHub
 - Bitbucket
 - ..
 - ..
4. your git repo will have multiple branches needed for end-to-end feature development.
5. master / main branch
6. integration / develop branch
7. for every new feature that you will code, you will cut a new feature branch from integration branch
8. clone the feature branch into your local machine. (sourcetree)

9. IN order to do the coding you need IDE + AIAgent + LLM help you in code

git => git branching

copy of code is branch.

=====

- There is a branching approach, which needs to be followed for application development.
- your MS git repo will have main branch, integration branch.
- for every new feature, you need to create a new feature branch from integration branch.
- clone this feature branch into your local machine.
- Import this code into you IDE. Devs will always code using IDE
 - AI Agent support for IDE is requirement
- Start coding in IDE.

=====

- While you are coding
 - RestAPI which has some business logic
 - for coding you will need framework knowledge - Spring & Spring Boot
 - Layered architecture approach while coding
 - controller - entry
 - how to do RestAPI dev.
 - Service

your functional coding in service

repository

working with DB

spring jdbc

spring data jpa (ORM) hibernate

modal

the data of your system will be represented as modal
pojo, dto, entities...

====

While you code:

- Know the frameworks - Spring / SpringBoot
- Coding as per layered architecture
 - for every layer, you need to learn how to do coding of that layer.

=====

COding

Testing in local machine

run this spring boot project in local machine

<http://localhost:8080/myapi>

browser

postman - API testing software

swagger

curl

Strong logging should be there in your application.

Logging: Slf4j with Logback

don't use `system.out.println`

After your functional coding is done, and local testing is done

Unittesting + Mocking + Code Coverage

code protects your code

you will write additional code which will check if your functional code is correct.

In AI time, its mandatory.

====

1. Code in local
2. Run spring boot app in local & do testing.
 API testing
 use postman, swagger, curl, browser
3. DO effective logging
 - Slf4j with logback
4. Complete Unittesting + Mocking + Code coverage
 JUnit Mockito ---

====

the code that you did in local, how does it reach the LIVE customer...

Code will be deployed in cloud - AWS, Azure, GCP,
LIVE customer can access it.

====

Code review
QA testing
UAT testing
Bug fixed & tested
GoLIVE - end customer

- your local should be moved to cloud feature branch (bitbucket)
 git add, git commit, git push

=> you need to deploy changes from feature branch into AWS dev environment.
 before giving to QA(Quality analyst) / testers, you want to validate that the feature is
working as expected.

=> raise a pull request from feature branch to integration branch
 include your reviewer for code review.

=> Lead do the code review, and then approve it, and merge your changes into integration

branch

=> all the feature which are being developed in that sprint, will be combined into integration branch.

=> integration branch will be deployed in QA env for tester testing.

2.5 years...

trainee

Week1

Mon-Fri, daily 3hrs LIVE on youtube

Daily tasks

Spring, Spring Boot

RestAPI

AWS deployment

Postman, Maven, IDEa.. coding

Git

Week1 & Week2

All core essential points needed for working on the project.

Week3 onwards - project functional development.

Sprint1: W3 & W4

Sprint2: W5 & W6

Sprint3: W7 & W8

100+ hrs of LIVE

Week2 on google meet.

Thursday

Project & joining for Week2.